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Paper 1: Machine Learning-Augmented Precision Dosing: Integrating Pharmacokinetic Modelin

Bibliographic Information

- **Title**: Machine Learning-Augmented Precision Dosing: Integrating Pharmacokinetic Modeling With Clinical Decision Support.
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Paper Type: GENERIC

Quick Take

This narrative review examines how machine learning can augment model-informed precision dosing (MIPD) across anti-infectives, triazole antifungals, and immunosuppressants. It argues that ML is most useful as a staged augmentation layer—triage, limited-sampling AUC estimation, residual correction, and workflow prioritization—while PopPK/Bayesian models and TDM remain the mechanistic and safety backbone. The review emphasizes external validation, uncertainty reporting, regulatory governance, and prospective outcome evidence as prerequisites for clinical adoption.

Executive Summary

The review synthesizes the intersection of ML, pharmacokinetic modeling, and clinical decision support for individualized dosing. It distinguishes direct ML prediction, hybrid pharmacometric-ML models, neural ODE/deep compartmental approaches, and reinforcement learning, and maps these to a priori and a posteriori decision stages. Across glycopeptides, β-lactams, polymyxins, triazoles, and immunosuppressants, the evidence is predominantly retrospective and heterogeneous; ML has shown promise for early exposure prediction, limited-sampling AUC estimation, toxicity risk stratification, and workflow triage, but few studies demonstrate improved patient outcomes. The authors propose a staged architecture in which EHR data and ML generate risk scores or preliminary estimates, TDM concentrations calibrate patient-specific predictions, PopPK/Bayesian simulation evaluates candidate regimens, and clinicians authorize doses with documented uncertainty and override pathways. They conclude that near-term adoption should integrate ML with established MIPD/TDM rather than replace it, and that external validation, transparent reporting (TRIPOD+AI, PROBAST+AI), privacy-preserving data infrastructure, and prospective multicenter trials are needed.

Scientific Context & Motivation

Inter-individual variability in drug exposure causes treatment failure or toxicity, especially in critically ill, older, organ-dysfunction, and polypharmacy patients. Routine MIPD relies on PopPK, Bayesian forecasting, and TDM; PBPK supports extrapolation and scenario simulation but is rarely used at the bedside. ML can capture nonlinear relationships in high-dimensional EHR/RWD, but its clinical role is unclear because predictive accuracy alone does not define actionability. The review addresses the gap between ML methodological innovation and clinical implementation by framing ML as an augmentation layer within MIPD, clarifying task-specific roles, data requirements, validation standards, and workflow integration.

⚡ Methodological Snapshot

This is a narrative review synthesizing recent methodological and clinical literature on ML applications in precision dosing. The authors structure the review around five dimensions: modeling strategies and integration pathways; data sourcing and feature engineering; model evaluation, reporting standards, and bias assessment; representative therapeutic domains; and translational challenges. They propose a taxonomy of ML strategies—direct prediction, limited-sampling AUC estimation, hybrid pharmacometric-ML models (parameter estimation, mechanistic feature enrichment/simulation augmentation, residual correction, embedded neural/continuous-time models), and reinforcement learning—and map these to a priori and a posteriori decision stages. The review includes two comparative tables (data sources and traditional TDM vs ML-assisted dosing) and a staged workflow figure. Evidence is drawn from studies in glycopeptides, β-lactams, polymyxins, triazole antifungals, and immunosuppressants, with emphasis on validation, reporting (TRIPOD+AI, PROBAST+AI, SPIRIT-AI, CONSORT-AI), governance, and implementation.

📋 Detailed Analysis

The review's central argument is that ML's clinical value is task-specific and depends on how it interfaces with pharmacological structure. Direct ML models can capture nonlinear covariate interactions and support a priori risk stratification, but they are vulnerable to distribution shift and may learn local practice patterns. Hybrid models preserve varying degrees of mechanistic constraint: ML-estimated parameters or features feed into PopPK/PKPD models; simulation augments sparse training data; residual learning corrects systematic pharmacometric error; and neural ODE/deep compartmental models embed dynamics within neural networks. Each category carries different extrapolation and interpretability trade-offs. The staged workflow is pragmatic: EHR data and ML triage before concentrations; TDM and Bayesian updating after concentrations; PopPK/Bayesian simulation to translate exposure estimates into candidate regimens; clinician authorization with uncertainty and override documentation. The review correctly notes that concentration forecasting is not equivalent to AUC estimation, and that many vancomycin ML studies target troughs rather than AUC-guided decisions. It also highlights that β-lactam MIPD trials have not uniformly improved outcomes, suggesting that risk stratification and targeted intervention may be more realistic than universal MIPD. For polymyxins, ML is positioned mainly for nephrotoxicity risk prediction rather than dose recalibration. For triazoles, ML integrates

pharmacogenomic and inflammatory covariates but transportability across ethnic groups is a concern. For immunosuppressants, limited-sampling AUC estimation with XGBoost has shown low bias in external full-PK datasets, but prospective outcome evidence is lacking. The review's limitations are those of a narrative review: no systematic search, no quantitative synthesis, and no formal risk-of-bias assessment. It also does not deeply address assay measurement error, model calibration drift, or the economics of implementation. Nevertheless, it provides a balanced roadmap for integrating ML into MIPD without compromising mechanistic safety.

Domain Context

Model-informed precision dosing (MIPD) combines population pharmacokinetic models, Bayesian forecasting, TDM, and therapeutic targets to individualize dosing. PBPK modeling complements MIPD for special populations, drug-drug interactions, and rare scenario simulation but is rarely used for bedside dose adjustment. ML has gained interest due to EHR/RWD availability, but clinical pharmacology has specific constraints: narrow therapeutic windows, exposure-response relationships, assay variability, sampling timing, and regulatory oversight. Reporting standards for AI in healthcare (TRIPOD+AI, PROBAST+AI, SPIRIT-AI, CONSORT-AI) are relevant. Data governance frameworks (GDPR, HIPAA) and privacy-preserving methods (federated learning, differential privacy) shape multicenter model development. The review sits at the intersection of pharmacometrics, clinical informatics, and implementation science, and it emphasizes that ML must respect pharmacological constraints and workflow realities.

☒ Key Findings

The review's central finding is that ML strategies must be matched to the clinical decision target and the availability of concentration data. Direct ML models can support a priori exposure prediction, limited-sampling AUC estimation, and outcome-risk stratification, but they are vulnerable to distribution shift and may learn local practice patterns. Hybrid pharmacometric-ML models preserve mechanistic constraints in four practical ways: ML estimation of covariate-parameter relationships; mechanistic feature enrichment or simulation-based data augmentation; residual correction of PopPK/Bayesian predictions; and embedded neural/continuous-time models such as neural ODEs and deep compartmental models. Each category carries different extrapolation, interpretability, and data requirements. Across therapeutic domains, evidence is most mature for glycopeptides and immunosuppressants. In vancomycin, ML studies often forecast trough concentrations rather than AUC, which may not directly support current AUC-guided practice; hybrid ML-PopPK approaches have reduced prediction error in sepsis, but prospective outcome data are limited. For β-lactams, randomized evidence suggests that unstratified MIPD does not consistently improve outcomes, so ML may be better positioned for probability-of-target-attainment modeling and early risk stratification. For polymyxins, ML is used mainly for nephrotoxicity risk prediction rather than dose recalibration. For triazoles, ML integrates pharmacogenomic and inflammatory covariates but transportability across ethnic groups is a concern. For tacrolimus and mycophenolic acid, XGBoost limited-sampling AUC models have shown low bias in external full-PK datasets, illustrating a concentration-based exposure estimator rather than autonomous

dose automation. The review also identifies workflow integration, external validation, uncertainty quantification, out-of-distribution checks, data governance, and regulatory classification as critical barriers. It proposes a staged architecture: EHR data and ML triage before concentrations; TDM and Bayesian updating after concentrations; PopPK/Bayesian simulation of candidate regimens; clinician authorization with documented uncertainty and override pathways; and longitudinal monitoring for changing physiology.

Strengths & Limitations

Strengths

- Comprehensive translational framing across modeling strategies, data infrastructure, validation, governance, and clinical domains.
- Clear taxonomy of hybrid ML-pharmacometric models: parameter estimation, mechanistic feature enrichment/simulation augmentation, residual correction, and embedded neural/continuous-time models.
- Distinguishes ML outputs by decision target (concentration vs AUC vs dose vs risk) and by a priori vs a posteriori stage, avoiding generic replacement-versus-augmentation claims.
- Emphasizes clinically meaningful endpoints, external validation, calibration, out-of-distribution checks, and reporting standards (TRIPOD+AI, PROBAST+AI, SPIRIT-AI, CONSORT-AI).
- Includes a practical staged workflow (Figure 1) and comparative tables across drug classes and data sources.
- Discusses data governance, privacy-preserving methods (federated learning, differential privacy), and regulatory classification.
- Balanced acknowledgment of evidence limitations and implementation barriers.

Limitations (Acknowledged by Authors)

- Most studies are retrospective, single-center, and lack prospective outcome data.
- Heterogeneity in dosing protocols, sampling, assays, and workflows limits transportability.
- Evidence maturity varies across drug classes; some domains remain exploratory.
- Reporting and validation standards are incompletely followed due to resource and infrastructure constraints.
- PBPK remains uncommon for bedside dose adjustment.
- ML models may learn local practice patterns and spurious correlations.
- Regulatory treatment of ML-based dosing software is uncertain and jurisdiction-dependent.

Limitations (Expert Review)

- Narrative review without a systematic search strategy, formal study selection, or quantitative synthesis; potential selection and interpretation bias.
- No formal risk-of-bias assessment of included studies (e.g., PROBAST) despite recommending it.
- Limited discussion of specific ML model architectures, hyperparameter tuning, sample-size requirements, and computational reproducibility.
- Sparse coverage of pediatric, obstetric, oncology, and gene-therapy dosing contexts; focus is on adult critical care and transplant populations.

- Does not quantify the added value of ML over Bayesian MIPD in terms of clinical outcomes or cost-effectiveness.
- Little discussion of assay standardization, limit of quantification, and measurement-error propagation into ML models.
- Reinforcement learning section is brief and does not deeply address reward specification, off-policy evaluation, or safety constraints.
- No patient/public involvement or implementation-science framework beyond workflow description.
- AI-assisted writing (OpenAI Codex) is disclosed, but its impact on reference checking and style is not independently evaluated.

Generalizability

The review's conceptual framework is broadly generalizable to precision dosing across therapeutic areas, but the evidence base is concentrated in adult critical care and transplant patients receiving narrow-therapeutic-index drugs. Findings may not directly transfer to pediatrics, pregnancy, oncology oral targeted agents, or resource-limited settings. The staged architecture is adaptable, but drug-specific validation, local workflow integration, and regulatory assessment are required before deployment.

Figures & Tables

- **Figure 1**: Staged integration of machine learning within model-informed precision dosing.
  - *Significance*: Operationalizes the proposed workflow: ML triage/prediction, TDM calibration, PopPK/Bayesian simulation, clinician authorization, and longitudinal monitoring; distinguishes PBPK's complementary role.
- **Table 1**: Data sources for ML-driven precision dosing.
  - *Significance*: Summarizes representative variables, clinical uses, bias risks, and implementation notes across EHR, labs, TDM, medications, ICU interventions, pharmacogenomics, registries, and simulated PK data.
- **Table 2**: Comparison of traditional TDM versus ML-assisted individualized dosing across therapeutic classes.
  - *Significance*: Contrasts conventional targets, clinical challenges, ML positioning, and evidence types for glycopeptides, β-lactams, polymyxins, triazoles, and immunosuppressants.

Code & Reproducibility Assessment

Not applicable — narrative review; no original code, software, or computational pipeline was released. The authors recommend reporting standards (TRIPOD+AI, PROBAST+AI) and versioned model monitoring for future ML-MIPD studies.

Future Directions

Prospective multicenter studies should compare clinically coherent strategies rather than isolated algorithms: ML limited-sampling AUC estimation versus Bayesian estimation; hybrid versus standalone models at distribution extremes; and workflow-integrated decision support versus usual care. Endpoints should include target attainment, toxicity, treatment failure, graft outcomes, alert burden, sampling use, turnaround time, override rates, and cost-effectiveness. Shared data models and federated learning infrastructure are needed to support multicenter development while respecting privacy. Neural ODE and deep compartmental models require qualification with uncertainty quantification and physiological constraints. Regulatory science for ML-based dosing software and implementation-science studies integrating ML into pharmacist-led workflows and EHRs are also priorities. Extension to pediatrics, pregnancy, oncology, and resource-limited settings is needed.

Expert Commentary

This is a well-timed, conceptually mature review that resists the temptation to overhype ML. Its main contribution is the staged augmentation architecture: ML earns a place only when it changes a defined decision, and concentration data plus mechanistic models remain the safety anchor. The taxonomy of hybrid models is useful for both pharmacometricians and clinical informaticists. However, the review is narrative rather than systematic, and the field still lacks prospective outcome trials. The most urgent next step is not a better algorithm but interoperable timestamped data, external validation, and workflow trials that measure patient outcomes. The authors' emphasis on uncertainty, out-of-distribution checks, and audit trails is exactly right for safety-critical dosing.

Bottom Line

For practitioners, ML should not be viewed as a standalone replacement for PopPK/Bayesian MIPD or TDM. Its most defensible near-term roles are triage (flagging patients for early TDM), limited-sampling AUC estimation with validated sampling windows, residual correction after Bayesian forecasting, and workflow prioritization. Any clinical deployment should pair ML outputs with concentration-based calibration, mechanistic simulation of candidate regimens, uncertainty/out-of-distribution warnings, clinician review, and audit trails. Adoption should await external validation and prospective evidence that the integrated pathway improves target attainment, toxicity, and resource use.

□ Figures

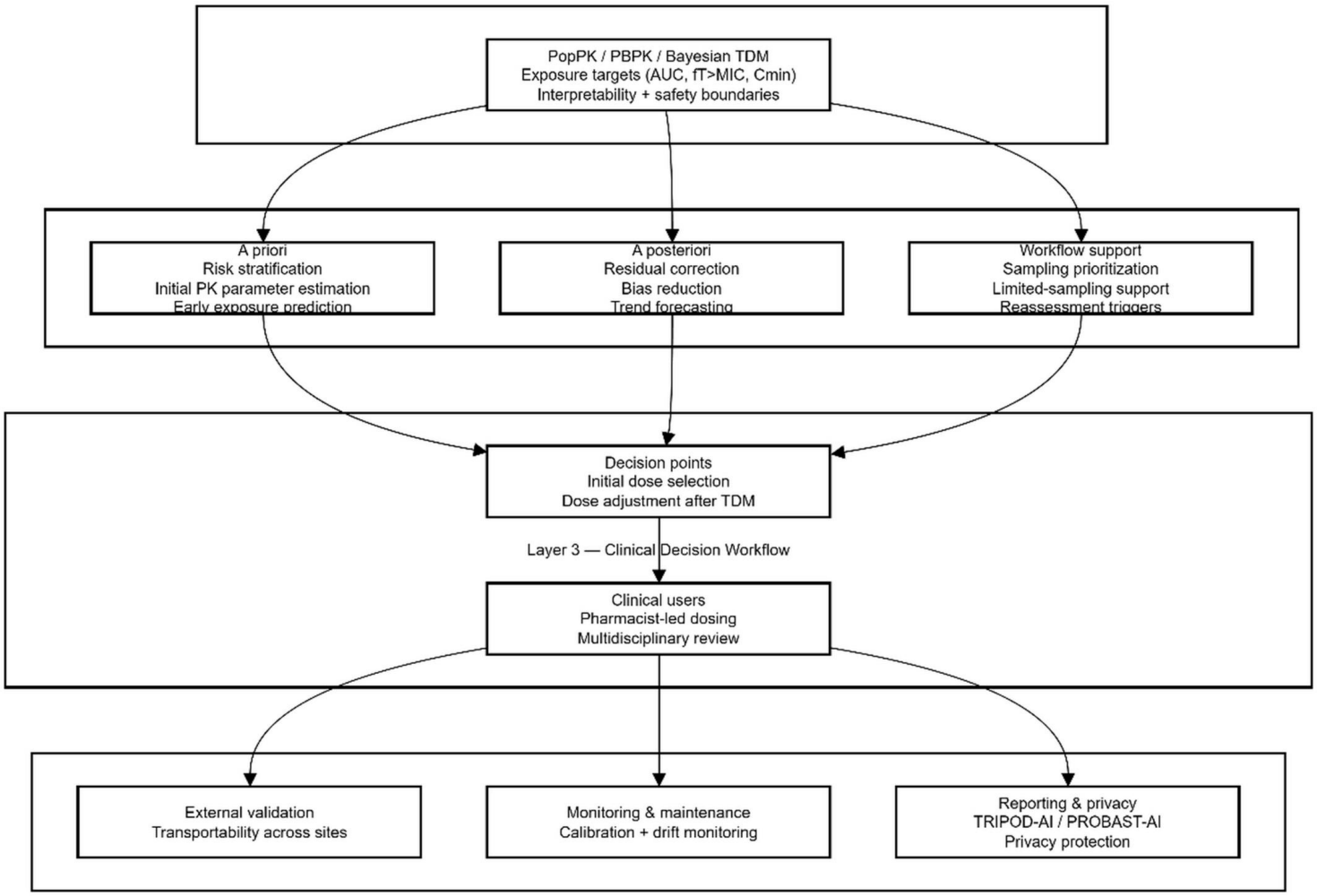


Figure 1: Staged integration of machine learning within model-informed precision dosing. Routine bedside MIPD is centered on PopPK/ Bayesian forecasting and TDM. ML may support a pr